

EFY Note

The source code of this project is available for free download at source.efymag.com

```
&off_btn, &on_btn1, &off_btn1, NULL);  
bool update_button(Adafruit_GFX_Button  
    *b, bool down)  
{  
    b->press(down && b->contains(pixel_x,  
        pixel_y));  
    if (b->justReleased())  
        b->drawButton(false);  
    if (b->justPressed())  
        b->drawButton(true);  
    return down;  
}  
bool update_button_list(Adafruit_GFX_  
    Button **pb)  
{  
    bool down = Touch_getXY();  
    for (int i = 0 ; pb[i] != NULL; i++) {  
        update_button(pb[i], down);  
    }  
    return down;  
}
```

In loop function, create several 'if conditions' that will check the status of the buttons and perform the tasks according to code.

Testing

After connecting the components, crosscheck the wirings and circuit connections. If all connections are alright, power on Arduino board with 5V DC to 12V DC—you can also use an AC-to-DC adaptor such as a cellphone charger or any 9V battery.

This touch-controlled multi-doorbell should be installed at the main gate of the building. Then extend the wires of doorbell speaker to the respective room or office cabin.

When you power on Arduino, you will get a menu with names of people in the rooms on the TFT screen. By default, the buttons and status on the screen will be blue. If you touch the button, say, office 1, it will become green and the doorbell will ring. Touch the name/office where you want to ring the bell. If you want to change the status, like office is closed or occupant is out of office, you can do it using Bluetooth module (HC-05). Simply connect the circuit to a suitable Bluetooth app on your smartphone through HC-05 module.

Once Bluetooth on the smartphone and HC-05 are connected, send commands set previously in the code to

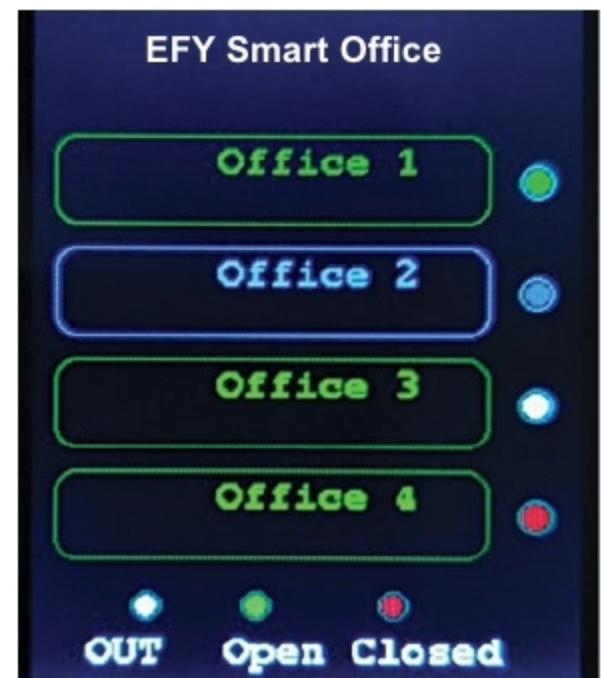


Fig. 6: Screenshot of the notification when someone is out of office room 3

change status. Screenshot of the notification when the person is out of office room 3, the status dot will turn white, as shown in Fig. 6. Bluetooth app is easily available on Google Play Store.

Note that, each piezo buzzer can be replaced with an actual doorbell along with a suitable driver circuit. **EFY**

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